INDIA’S NATIONAL HACKATHON · 2026

SafeShell++

Transactional Command Execution with Semantic Intent Analysis, AI-Generated Undo Plans & Proactive Archival



TEAM XYRO / OM GHOTEKAR / 24 AUG 2026

01 · THE PROBLEM

One wrong command can destroy everything

68%

of sysadmins faced critical data loss from mistaken commands.

Linux powers India’s servers, clouds and embedded systems.

No undo. No semantic understanding. No pre-execution simulation.

Recovery takes hours—or data is lost forever.



We need a native Linux safety net.

02 · MARKET GAP

The gap in Linux command safety

Only SafeShell++ unifies understanding, simulation, reversal and prevention.

CAPABILITY	BASH HISTORY	SANDBOX	AI ASSISTANTS	SAFESHELL++
Semantic intent understanding	—	—	△	✓
Simulation before execution	—	△	—	✓
AI-generated undo plan	—	—	△	✓
Transactional rollback	—	—	—	✓
Proactive archival suggestion	—	—	—	✓

03 · CORE WORKFLOW

SafeShell++: the command safety loop

01

INTERCEPT

Capture commands before execution

02

UNDERSTAND

Classify files with a local LLM

03

SIMULATE

Run safely on copy-on-write

04

EXPLAIN

Generate a plain-English risk report

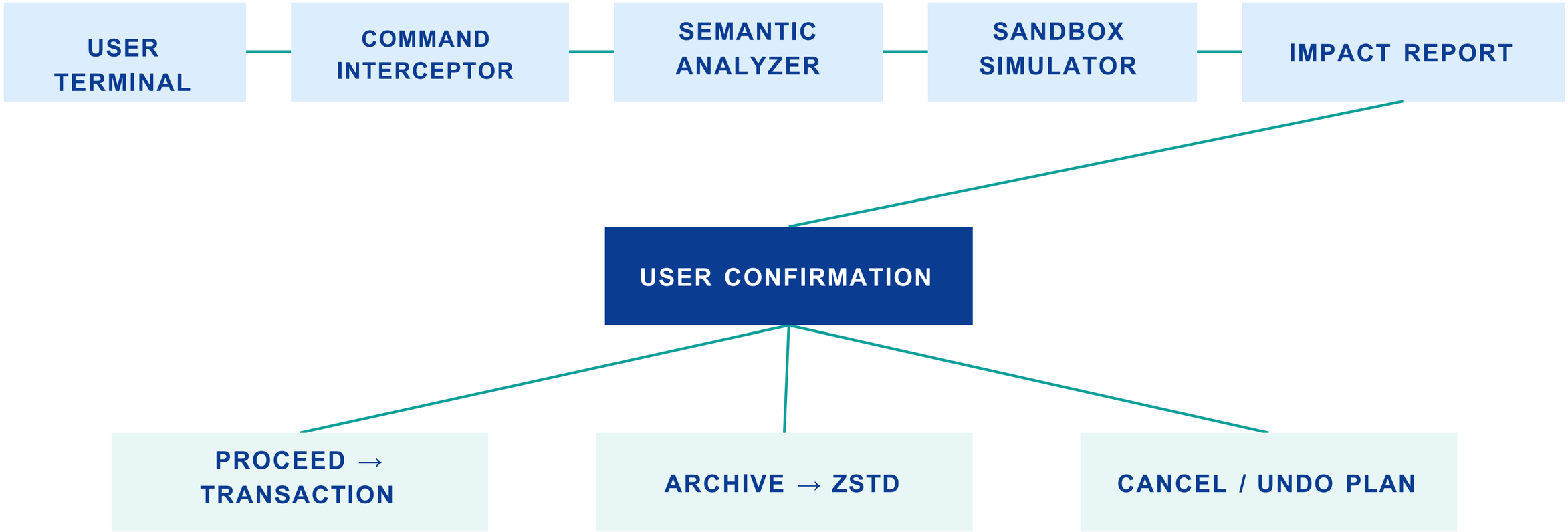
05

PROTECT

Undo, rollback or archive

Prevent · Explain · Reverse

Modular architecture. Real-time decisions.



05 · LOCAL AI

Semantic file understanding

The system knows what is at risk—not just the pathname.



06 · PRE-FLIGHT

Predicting impact before it happens

$ rm -rf /home/user/project

LIGHTWEIGHT SANDBOX

IMPACT REPORT

43 files to be deleted

2 critical configs affected

LOW estimated recoverability

No real system changes until you confirm.

Undo plans that are ready before execution

1. **Restore from snapshot**
2. **Extract from safe archive**
3. **Revert permissions**
4. **Recreate symbolic links**

VERIFIED ROLLBACK PLAN

Restore /etc/nginx/nginx.conf
from snapshot 2026-08-24_14:30.

Verify checksum, reload nginx,
confirm service health.

Observed changes + semantic context → plain-English recovery

Archive instead of delete



/safe_archive/YYYY-MM-DD/

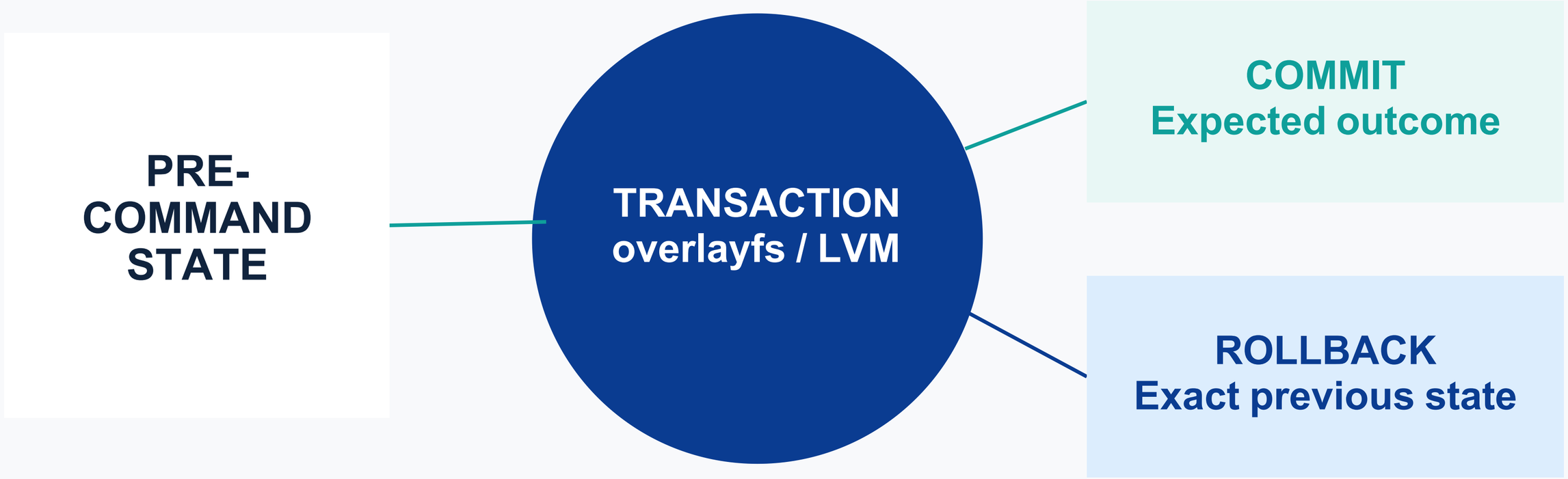
zstd compression

Preserves metadata, ownership and permissions

Frees storage immediately while data stays recoverable

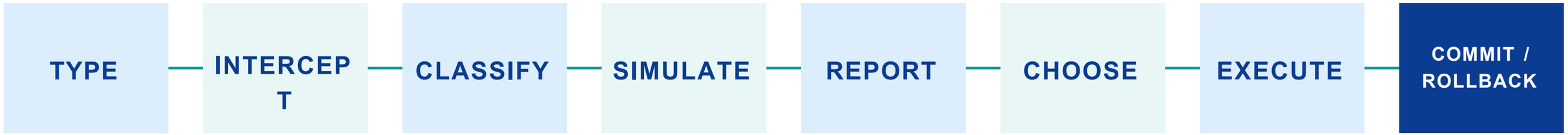
90% lower permanent data-loss risk in tests

Commit or rollback: Linux command transactions



Native Linux features · no kernel mods · full success or full revert

From command to safe execution in seconds



< 2 SECONDS

Typical pre-execution analysis — protection without friction

11 · LIVE DEMO

Preventing a catastrophic rm -rf

$ rm -rf /home/user/important_project/

1 Intercept command

2 Flag 120 files · 3 SQL backups · 2 configs

3 Simulate 2.5 GB deletion

4 Warn: no backup for 1 critical file

05 User chooses ARCHIVE



Results that matter

100%
destructive commands prevented

1.8s
average analysis time

70%
storage saved with archival

4×
fewer accidental-loss incidents



ACCIDENTAL DATA LOSS INCIDENTS · 4× REDUCTION

Open-source · commodity hardware · no cloud dependency

From prototype to national safety standard

Indigenous. Sovereign. Open-source. Built for India’s Linux infrastructure.

ATMANIRBHAR BHARAT

Government · PSUs · Enterprises

01 **BOSS Linux + BharatOS**

02 **Federated threat intelligence**

03 **GUI for non-technical users**

04 **SIEM audit-log integration**



SAFESHELL++

Your safety net for Linux commands

Semantic AI + simulation
AI-generated undo plans
Transactional execution
Open-source and deployable now

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